



An Internship Report
Of
Data Analyst

Submitted To
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IILM University, Greater Noida, U.P

In partial fulfilment of the requirement of degree of
B.Tech (CSE)

BY

Roll Number Of Student :- 25SCS1003005621

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Batch :- 2026-27

CANDIDATE'S DECLARATION

I, KHUSHI RAI, do hereby declare that the internship report titled "Internship Report on Data Analytics" has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science Engineering. All the references have been duly acknowledged, and I have not incorporated any material from any other source without citation. I affirm that this report is entirely my own work and has not been submitted, in full or in part, to any other institute for the award of any degree or diploma, and I shall be solely responsible for any copyright violation arising in this regard.

Signature

Student Name: KHUSHI RAI

Roll No.: 25SCS1003005621

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to BeeSkilled for giving me the opportunity to undertake this six-week internship in the Data Analyst domain. This internship exposed me to practical, real-world data analysis practices and allowed me to apply the concepts learned during my academic curriculum in a hands-on setting.

I am thankful to IILM University, Greater Noida, U.P. and the School of Computer Science and Engineering for providing the academic foundation and encouragement that made this internship possible. I extend my heartfelt thanks to my faculty members and academic support staff for their guidance, constructive feedback, and continued support throughout this internship period.

I am also grateful to everyone at BeeSkilled who supported me during this internship for sharing their knowledge and insights on various aspects of Data Analytics. Finally, I express my deepest gratitude to my family for their constant encouragement and support throughout this journey.

Signature

Student Name: KHUSHI RAI
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The certificate below was issued by BeeSkilled on 28 August 2026, confirming successful completion of the 6-week online internship program in the Data Analyst domain.



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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

Data Analytics refers to the systematic process of studying and interpreting data to uncover useful patterns, evaluate performance, and support effective decision-making. With organizations increasingly depending on data to guide their operations and strategies, the ability to organize, analyze, and interpret information has become an essential part of modern business practices. The field combines a variety of techniques, ranging from spreadsheet-based calculations and statistical analysis to programmatic data processing and interactive business intelligence reporting.

An important part of the analytical process is preparing the data before performing detailed analysis. Datasets may contain missing entries, duplicate records, inconsistent formats, or incorrectly structured values that can affect the accuracy of subsequent results. Data cleaning and preparation therefore play a crucial role in ensuring that the information being analyzed is reliable. Once the data has been prepared, tools such as Microsoft Excel, Python with Pandas, and Power BI can be used to explore datasets, perform calculations, develop business measures, visualize trends, and communicate findings effectively.

This report presents the work carried out during a six-week internship in the Data Analyst domain at BeeSkilled, with the practical activities documented across four major learning modules. The internship covered Excel-based sales analysis, Python and Pandas dataset exploration, Python-based data cleaning and wrangling, and an end-to-end business analytics and visualization capstone project. The overall experience provided an opportunity to apply concepts learned through the B.Tech CSE curriculum to practical data-analysis tasks and understand how raw datasets can be systematically processed and converted into meaningful, decision-oriented business insights.

4.2 Organization Profile

BeeSkilled is an online platform that provides internship and skill-development programs under the tagline “Tech and Internships.” Its programs are structured around practical projects and assignments, with the objective of helping participants strengthen their technical knowledge and gain hands-on exposure to industry-oriented practices. The organization is registered with AICTE (All India Council for Technical Education) and MSME (Ministry of Micro, Small and Medium Enterprises), as mentioned in the official internship communication.

The internship described in this report was completed in the Data Analyst domain as part of a six-week online program. The program combined weekly learning modules with practical assignments and an end-to-end capstone project, enabling the concepts covered during the training to be applied to different datasets and analytical tasks. The program was conducted under the guidance of Ms. Simranjeet Kaur, Founder & Program Head. Upon successful completion of the prescribed coursework and capstone project, an internship completion certificate was issued by BeeSkilled.

4.3 Problem Statement

Organizations collect substantial amounts of raw information through their routine business activities, yet such data becomes valuable only after it has been processed and interpreted to support meaningful decision-making. The central challenge addressed during this internship was understanding how raw, incomplete, or unorganized datasets can be transformed into useful and actionable business insights through a systematic analytical workflow:

**Raw Data → Data Cleaning → Data Transformation → DataAnalysis → Visualization →
Insights → Recommendations**

This workflow was implemented through practical activities throughout the internship: in Excel, by organizing and summarizing a large orders dataset using Pivot Tables and business metrics; in Python and Pandas, by examining a transactional dataset and independently cleaning an exercise-tracking dataset containing missing and duplicate records; and during the Week 4 capstone project, by processing a raw sales-transaction dataset through cleaning, validation, exploratory analysis, and business evaluation to generate meaningful findings and stakeholder-focused recommendations.

4.4 Project Objectives

- Develop a strong understanding of Data Analytics and its importance in supporting effective business decision-making.
- Gain practical knowledge of Excel-based data analysis, including Pivot Tables and aggregation techniques for large transactional datasets.
- Examine revenue, sales, and category-level performance using Pivot Table-based analytical summaries and dashboards.
- Build a foundation in Python programming and apply the Pandas library for systematic exploration of structured datasets.

- Apply fundamental Pandas functions, including head(), info(), shape, and describe(), to examine and summarize dataset characteristics.
- Perform data-cleaning procedures by detecting and appropriately treating missing values and duplicate records within datasets.
- Transform irregular date fields into standardized datetime formats and apply suitable methods for handling missing entries.
- Generate derived and calculated columns from existing attributes to support additional analysis.
- Apply condition-based filtering to extract records that satisfy specific analytical or business requirements.
- Gain practical understanding of Power BI dashboard development and the creation of calculated measures using DAX.
- Develop data storytelling skills by organizing analytical findings into clear and meaningful information for stakeholders.
- Implement a complete data analytics workflow through the capstone project, progressing from raw data preparation and analysis to meaningful business recommendations.

4.5 Scope of the Project

The internship primarily focused on structured, dataset-oriented analysis using sample and practice datasets included in the BeeSkilled curriculum. The activities were carried out in an academic and training environment and did not involve working with live production systems or confidential organizational datasets. Within this scope, the internship included:

- Managing and organizing structured business datasets, particularly sales and order-related information.
- Performing Pivot Table-based analysis and summarizing important business metrics using Microsoft Excel.
- Examining and exploring datasets with Python and the Pandas library to understand their structure and characteristics.
- Cleaning and transforming data, including the treatment of missing values, removal of duplicate records, and correction of inconsistent date formats.
- Creating derived variables and applying condition-based filters to extract useful information from datasets.

- Validating data accuracy and consistency by checking relationships between related numerical fields.
- Developing business intelligence measures using DAX and understanding dashboard creation concepts in Power BI.
- Applying data storytelling techniques to convert analytical findings into clear insights and practical recommendations for stakeholders.

4.6 Technologies and Tools Used

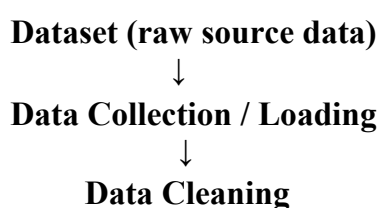
Table 4.1 summarizes the primary tools and technologies used during the internship, along with their purpose and how each was applied.

Tool	Purpose	Application during Internship
Microsoft Excel	Spreadsheet-based data organization and analysis	Built a pivot-table dashboard analyzing total revenue, sales by category, yearly trends, and segment-wise revenue from a 51,290-record orders dataset
Python	General-purpose programming for data analysis tasks	Used for dataset loading, inspection, and data-cleaning scripts across Weeks 2 and 3
Pandas	Data manipulation — reading, inspecting, cleaning, and transforming data	Inspected a 200-row sales dataset using <code>head()/info()/describe()</code> ; cleaned a 32-row exercise dataset by handling missing values, duplicates, and date formatting; created a derived column and filtered records
Power BI	Business intelligence dashboarding and calculated (DAX) measures	Built DAX measures — Total Sales, Total Profit, Total Units Sold, Profit Margin, Average Order Value, and Loss-Making Orders — for the Week 4 capstone dashboard

Table 4.1: Technologies and Tools Used

4.7 Data Analytics Workflow

Rather than a software-engineering system architecture, the internship followed a Data Analytics workflow that moves from raw data collection through cleaning, transformation, and analysis, to visualization and, finally, actionable recommendations. This workflow, shown in Figure 4.1, was applied consistently across all four weeks of the internship.



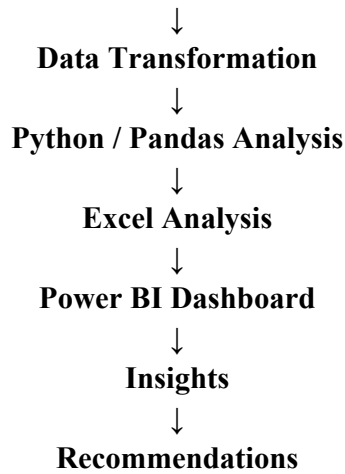


Figure 4.1: Data Analytics Workflow

4.8 Methodology

This section documents the methodology followed during each week of the internship, along with the specific work performed and the results obtained.

4.8.1 Week 1: Excel Data Analysis

During the first week of the internship, the focus was on developing fundamental Excel-based data analysis skills, including spreadsheet operations, aggregation formulas, Pivot Tables, and methods for organizing and summarizing large datasets. The practical assignment required analyzing sales performance using the “Global Superstore” Orders dataset, which consisted of approximately 51,290 records distributed across 24 columns. As illustrated in Figure 4.2, the category-level analysis identified Technology as the leading category with sales of approximately \$4.74M, followed by Furniture at around \$4.11M and Office Supplies at approximately \$3.79M. The yearly analysis also indicated a steady growth in sales, increasing from approximately \$2.26M in 2012 to nearly \$4.30M in 2015.

<i>Sales by Category</i>	<i>Total Sales</i>
Technology	\$4.74M
Furniture	\$4.11M
Office Supplies	\$3.79M

<i>Yearly Sales Trend</i>	<i>Total Sales</i>
2012	\$2.26M

<i>Yearly Sales Trend</i>	<i>Total Sales</i>
2013	\$2.68M
2014	\$3.41M
2015	\$4.30M

Figure 4.2: Excel Pivot Table — Sales Performance Analysis (Category-wise and Yearly)

Customer-segment-wise revenue was also analyzed using pivot tables, shown in Figure 4.3. The Consumer segment generated the highest revenue at approximately \$6.51M, followed by Corporate at approximately \$3.82M and Home Office at approximately \$2.31M.

<i>Customer Segment</i>	<i>Total Sales</i>
Consumer	\$6.51M
Corporate	\$3.82M
Home Office	\$2.31M

Figure 4.3: Segment-wise Revenue Analysis

This exercise demonstrated how Excel pivot tables can be used to organize a large raw transactional dataset into summarized, decision-ready business metrics using SUM-based aggregation, without writing any code.

4.8.2 Week 2: Python/Pandas Data Analysis

The second week of the internship concentrated on using Python and the Pandas library for structured data analysis. The learning activities covered importing datasets through Pandas, examining their structure and data types, and obtaining descriptive statistical information. The practical task for this week involved loading a sales dataset and carrying out an initial examination of its contents.

A sales dataset comprising 200 records and 10 columns was imported using Pandas. The dataset contained the fields `order_id`, `customer_name`, `order_date`, `category`, `sub_category`, `product_name`, `quantity`, `unit_price`, `total_price`, and `region`. Its dimensions and column data types were examined using `df.shape` and `df.info()`, respectively, as demonstrated in Figure 4.4.

```
>>> df.shape
(200, 10)

>>> df.info()
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
```

```
Data columns (total 10 columns):
#      Column      Non-Null Count  Dtype
---  -
0      order_id      200      non-null  int64
1      customer_name  200      non-null  object
2      order_date     200      non-null  object
3      category       200      non-null  object
4      sub_category   200      non-null  object
5      product_name   200      non-null  object
6      quantity       200      non-null  int64
7      unit_price     200      non-null  float64
8      total_price    200      non-null  float64
9      region        200      non-null  object
```

Figure 4.4: Dataset Structure Summary (df.shape and df.info() Output)

Figure 4.5 shows the first five records of the dataset retrieved using the `df.head()` method, giving an initial view of the raw transactional data prior to analysis, along with the list of column names retrieved using `df.columns`.

```
>>> df.columns.tolist()
['order_id', 'customer_name', 'order_date', 'category',
 'sub_category', 'product_name', 'quantity', 'unit_price',
 'total_price', 'region']

>>> df.head()
   order_id customer_name order_date      category  quantity
0      1000      <customer>      <date>      <category>         ..
1      1001      <customer>      <date>      <category>         ..
2      1002      <customer>      <date>      <category>         ..
3      1003      <customer>      <date>      <category>         ..
4      1004      <customer>      <date>      <category>         ..
```

Figure 4.5: Sample Dataset Records (df.head() and df.columns Output)

Figure 4.6 presents the descriptive statistical summary obtained through the `df.describe()` method for the numerical fields — `quantity`, `unit_price`, and `total_price` — providing a concise view of the dataset's overall distribution and central values. The analysis showed an average quantity of 4.91 units per order, while the average unit price was 2421.495 and the average total price reached 12,101.535. The dataset also recorded a maximum total price of 47,940, representing the highest transaction value observed.

```
>>> df.describe()
      quantity      unit_price      total_price
count  200.000000    200.000000    200.000000
mean     4.910000    2421.495000    12101.535000
std       ...           ...           ...
min       ...           ...           ...
max       ...           ...    47940.000000
```

Figure 4.6: Statistical Summary of the Dataset (df.describe() Output)

The exercise demonstrated that using `shape`, `info()`, `head()`, and `describe()` together provides an efficient initial understanding of an unfamiliar dataset. These functions help determine its dimensions, examine column types, preview sample records, and assess numerical distributions before proceeding with more detailed analysis.

4.8.3 Week 3: Python and Data Wrangling

The third week of the internship emphasized data wrangling with Python and Pandas, with particular attention to recognizing and addressing common data-quality problems. The activities involved working with missing information, duplicate entries, irregular date formats, and the creation of additional calculated fields. The primary practical task was to clean the given dataset using Pandas and generate a processed version for subsequent analysis.

For this exercise, a workout and activity dataset containing 32 records and 5 columns — Duration, Date, Pulse, Maxpulse, and Calories — was imported and examined. The initial assessment identified one duplicate record, one missing value in the Date field, and two missing entries in the Calories field, as illustrated in Figure 4.7.

```
Original dataset      : 32 rows x 5 columns
Columns              : Duration, Date, Pulse, Maxpulse, Calories

Data quality issues found:
- Missing Date values      : 1
- Missing Calories values : 2
- Duplicate rows           : 1

Cleaning actions performed:
- Removed 1 duplicate row
- Converted Date column to proper datetime format
- Filled missing Date value using interpolation
- Filled missing Calories values using the column mean
- Added derived column: Calories_Per_Minute
  Calories_Per_Minute = Calories / Duration

Cleaned dataset      : 31 rows x 6 columns
```

Figure 4.7: Data Cleaning / Transformation Summary

Following the data-preparation process, the Date column, which was initially represented in an inconsistent text format, was transformed into a standardized datetime format. The single missing Date value was addressed through interpolation, while the two missing Calories entries were replaced with the mean value of the column. After these corrections, the dataset contained no remaining missing values. A new calculated field, `Calories_Per_Minute`, was generated by dividing Calories by Duration for each record. With the duplicate removed and the derived column added, the processed dataset contained 31 rows and 6 columns and was exported as `cleaned_data.csv`.

As an additional task, the cleaned dataset was filtered to include only records having Calories greater than 300, and this resulting subset was saved separately as `filtered_data.csv`. The exercise emphasized the significance of data preparation prior to analysis, as missing values, duplicate entries, and inconsistent formats may influence subsequent calculations and analytical results. It also reinforced the need to verify data quality before carrying out further transformations, filtering, or interpretation.

4.8.4 Week 4: Data Visualization and Capstone Project

The fourth week focused on Power BI dashboard development, with emphasis on creating calculated DAX measures and presenting important business metrics through visualizations. The module also introduced data storytelling, particularly the process of organizing analytical results into meaningful insights and practical recommendations for stakeholders. The week concluded with an End-to-End Business Analytics Capstone Project, where the complete data analytics workflow was applied to a financial and sales transaction dataset.

The capstone dataset consisted of 700 records and 16 original columns, including Segment, Country, Product, Discount Band, Units Sold, Gross Sales, Discounts, Sales, COGS, Profit, Date, Month, and Year.

During the data-cleaning and validation stage, 53 records were identified with blank values in the Discount Band field. Further investigation showed that these records had zero discount amounts, so they were categorized as “No Discount” instead of being removed, thereby retaining the valid transactions. The dataset contained no duplicate records. Its numerical consistency was also verified across all 700 rows by confirming that $\text{Units Sold} \times \text{Sale Price} = \text{Gross Sales}$, $\text{Gross Sales} - \text{Discounts} = \text{Sales}$, and $\text{Sales} - \text{COGS} = \text{Profit}$. A Year-Month helper field was created to facilitate monthly trend analysis, and the cleaned data was exported as `Week_4_Cleaned_Data.csv`. After completing the cleaning and validation process, exploratory business analysis was performed and a Power BI dashboard with calculated DAX measures was developed, as presented in Figure 4.8.

<i>Business KPI</i>	<i>Value</i>
Total Sales	\$118,726,350
Total Profit	\$16,893,702
Total Units Sold	1,125,806
Average Order Value	\$169,609
Profit Margin	14.2%
Total Orders	700

Figure 4.8: Power BI Business Performance Dashboard — Key KPI Measures

The dashboard incorporated key DAX measures, including Total Sales, Total Profit, Total Units Sold, Profit Margin, Average Order Value, and Loss-Making Orders. The major insights obtained from the exploratory analysis and Power BI dashboard were as follows:

- Government emerged as the highest-profit segment, generating approximately \$11.39M, while Enterprise was the only segment showing an overall loss of approximately -\$614,546.
- The USA achieved the highest sales among the countries analyzed, recording approximately \$25.03M in revenue.
- Paseo delivered the strongest performance in terms of both sales and profit, whereas Carretera recorded the lowest performance across both measures.
- 58 out of 700 orders, representing approximately 8.3%, were identified as loss-making, with these orders mainly linked to the Medium and High discount bands.
- The monthly analysis showed October and December 2014 among the strongest sales periods, while September 2013 recorded the lowest sales during the analyzed timeframe.

Based on the insights obtained from the analysis, the following recommendations were formulated to support stakeholder decision-making:

- Reassess discount-approval procedures, particularly for higher discount levels, due to their observed relationship with loss-making transactions.
- Establish minimum profit-margin thresholds for heavily discounted orders to minimize the possibility of accepting unprofitable deals.
- Examine the factors contributing to the Enterprise segment's overall loss before making decisions regarding further expansion or investment in the segment.
- Evaluate Carretera's pricing strategy and cost structure, considering its comparatively weak performance in both sales and profitability.
- Align inventory, staffing, and other resources with seasonal demand patterns, particularly the higher sales activity observed during October–December.
- Investigate the unusually low sales recorded in September 2013 to determine whether specific underlying factors contributed to the decline and identify suitable corrective actions.

The capstone project integrated the key skills acquired throughout the internship, including data cleaning, data validation, exploratory analysis, and Power BI measure development, into a unified end-to-end analytical process. It demonstrated the complete progression from preparing raw transactional data to generating meaningful insights and formulating clear, stakeholder-oriented business recommendations.

4.9 Expected Outcomes / Learning Outcomes

- Practical proficiency in Excel-based data analysis, including the creation of Pivot Tables and summarization of key business metrics from large datasets.
- Working knowledge of Python programming and Pandas, with experience in inspecting, exploring, and understanding structured datasets.
- Hands-on exposure to data-cleaning techniques, including detecting and treating missing values, removing duplicate records, and correcting inconsistent date formats.
- Ability to construct derived or calculated columns and perform condition-based filtering for extracting relevant dataset records.
- Familiarity with Power BI dashboard development and the implementation of calculated business measures using DAX.
- Experience applying a complete analytics workflow to transactional data, covering data cleaning, validation, exploratory analysis, KPI evaluation, and reporting.
- Enhanced ability to interpret analytical results and communicate them through concise, actionable, and stakeholder-focused business recommendations.

4.10 Certificates of Completion and Other Work Proof

The Internship Completion Certificate issued by BeeSkilled, confirming successful completion of the 6-week Data Analyst internship, is presented in the preliminary pages of this report (page iii). Figure 4.9 below reproduces the original internship offer letter issued at the start of the program as supporting communication proof.



Figure 4.9: BeeSkilled Internship Offer Letter

No additional communication or completion emails beyond the documents shown above were available at the time of preparing this report.

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

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- BeeSkilled. Data Analyst Internship Curriculum Materials — Week 1 to Week 4, provided as part of the internship training program.